

A Variational Bound on Generalization Error for Stochastic Neural Networks

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Abstract

We derive a PAC-Bayesian bound on the generalization error of stochastic neural networks. The bound is expressed in terms of the KL divergence between the posterior and prior weight distributions and the training-set size.

1 Setup

Let $\mathcal{D}$ be a data distribution over input-label pairs, and let $h_w : \mathcal{X} \rightarrow \mathcal{Y}$ be a classifier parameterized by weights $w \sim Q$ drawn from a posterior Q . Let P be a data-independent prior over weights. For a bounded loss $\ell \in [0, 1]$:

$$L(Q) = \mathbb{E}_{w \sim Q} \mathbb{E}_{(x,y) \sim \mathcal{D}} \ell(h_w(x), y) \quad (\text{true risk})$$

$$\hat{L}_n(Q) = \mathbb{E}_{w \sim Q} \frac{1}{n} \sum_{i=1}^n \ell(h_w(x_i), y_i) \quad (\text{empirical risk})$$

2 Main Theorem

For any posterior Q , prior P independent of the training data, sample size n , and $\delta \in (0, 1)$, with probability at least $1 - \delta$:

$$L(Q) \leq \hat{L}_n(Q) + \sqrt{\frac{\text{KL}(Q \parallel P) + \log(2\sqrt{n}/\delta)}{2n}} \quad (1)$$

3 Asymptotic Behavior

As $n \rightarrow \infty$, the generalization gap vanishes at rate:

$$L(Q) - \hat{L}_n(Q) = O\left(\sqrt{\frac{\text{KL}(Q \parallel P)}{n}}\right)$$

4 Specialization to Gaussian Posteriors

When $Q = \mathcal{N}(\mu_Q, \Sigma_Q)$ and $P = \mathcal{N}(0, \sigma^2 I)$, the KL has closed form:

$$\text{KL}(Q \parallel P) = \frac{1}{2} (\text{tr}(\Sigma_Q/\sigma^2) + \|\mu_Q\|^2/\sigma^2 - d + d \log \sigma^2 - \log |\Sigma_Q|) \quad (2)$$

where d is the parameter dimension.

5 Discussion

The bound is non-vacuous for wide networks when the KL can be controlled via early stopping or explicit regularization.